Bosonization of Majorana CFT with Boundaries

Yu Liu

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Abstract

It is well known that the two-dimensional critical Ising model can be described by a free massless Majorana fermion CFT as well as the $c = \frac{1}{2}$ minimal model Ising CFT. It is interesting to show the explicit correspondence between the two theories.

In this note, we focus on how this two theories are related with the presence of boundaries. We start from the Majorana CFT with several types of boundary conditions, and we calculate the partition functions on a compactified strip or equivalently a cylinder in both open and closed channels. Then we use the closed channel to bosonize the Majorana CFT, which indeed gives us the boundary states and partition functions of the Ising BCFT.

Contents

1 Preliminaries

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1.1 Boundary CFT

I will first give a very brief review of fundamental concepts and tools in Boundary Conformal Field Theory (BCFT). For a more detailed introduction, one can refer to my CFT and BCFT note, click here, and the following references [1, 7, 8].

1.1.1 Gluing Condition

A Boundary CFT (BCFT) is a conformal field theory on a manifold with boundary induced from a parent CFT. It inherets the operator algebra data of the parent CFT, while having extra data due to the presence of boundaries.

The simplest case is to consider a BCFT on the upper half plane (UHP) $\{z = x + iy | y \geq 0\}$, where the real axis $y = 0$ serves as the boundary. To preserve conformal symmetry at the boundary, we need to impose the gluing condition on the energy-momentum tensor $T(z)$ and $\bar{T}(\bar{z})$ at the boundary:

Definition 1. ***Gluing Condition***
At the boundary $y = 0$, the energy-momentum tensor satisfies:

$$T(z) = \bar{T}(\bar{z}) \quad \textit{at } y = 0. \tag{1}$$

With this gluing condition, we can prove that the conformal symmetry is ”partially” preserved, which leads to the existence of a single copy of Virasoro algebra as symmetry algebra instead of two:

Theorem 1. ***Symmetry Algebra of BCFT***
In a BCFT defined on the upper half plane with the gluing condition $T(z) = \bar{T}(\bar{z})$ at the boundary $y = 0$, the symmetry algebra is generated by a single Virasoro algebra with generators:

$$L_n^H = \frac{1}{2\pi i} \oint_C dz\, z^{n+1} T^H(z) \tag{2}$$

where $T^H(z)$ is the analytical continuation of $T(z)$ and $\bar{T}(\bar{z})$. The generators satisfy:

$$[L_n^H, L_m^H] = (n-m) L_{n+m}^H + \frac{c}{12} n(n^2-1)\delta_{n+m,0} \tag{3}$$

1.1.2 Boundary States Formalism

There exist various ways of defining a BCFT from a parent CFT on a manifold by imposing different types of boundary conditions at the boundary. A tool of labeling and classifying these boundary conditions is the boundary state formalism. We define the boundary states as;

Definition 2. ***Boundary State***

Boundary states $|\alpha\rangle$ and $|\beta\rangle$ are defined as states living on the boundary of the parent CFT placed on a chopped cylinder, and they satisfy:

- *The cylinder has the same geometry as a compactified BCFT on a strip, with length L and circumference β.*
- *The partition function of the parent CFT on a cylinder with boundary states $|\alpha\rangle$ and $|\beta\rangle$ at its two ends is equivalent to the partition function of the BCFT on a strip with boundary conditions α and β at the two ends. Mathematically:*

$$Z_{\alpha\beta} = \langle\alpha| \, \tilde{q}^{\frac{1}{2}(L_0+\bar{L}_0-c/12)} \, |\beta\rangle = \mathrm{Tr}_{\mathcal{H}_{\alpha\beta}} \left(q^{L_0^H - c/24} \right), \tag{4}$$

where $q = e^{-\pi\beta/L}$ and $\tilde{q} = e^{-4\pi L/\beta}$.Here L_0 and $\bar{L}_0$ are the Virasoro generators of the parent CFT, while L_0^H is the Virasoro generator in the BCFT. These condition are often referred to the Cardy's condition.

We can use the boundary states as a state in the Hilbert space of the parent CFT to encode the boundary conditions of the BCFT. To be consistent with the gluing condition, the boundary states must satisfy the following Ishibashi condition:

Theorem 2. ***Ishibashi Condition***

Boundary states $|\alpha\rangle$ should satisfy:

$$(L_n - \bar{L}_{-n}) \, |\alpha\rangle = 0, \quad \forall n \in \mathbb{Z}. \tag{5}$$

A solution to the Ishibashi condition can be constructed from each primary state $|h\rangle$ of the parent CFT, leading to the definition of Ishibashi states:

Definition 3. ***Ishibashi State***

For each primary state $|h\rangle$ of the parent CFT, the corresponding Ishibashi state $|h\rangle\rangle$ is defined as:

$$|h\rangle\rangle = \sum_N |h; N\rangle \otimes \overline{|h; N\rangle}, \tag{6}$$

where $\{|h; N\rangle\}$ is an orthonormal basis of the Verma module built on the chiral primary state $|h\rangle$.

1.1.3 Cardy's Condition

Consider the parent CFT to be a diagonal CFT. Indeed, the main subject of this note, the Ising CFT, is a diagonal CFT. There exists an equivalent formulation of Cardy's condition:

Theorem 3. ***Cardy's Condition****(Diagonal CFT)*

For two boundary states $|\alpha\rangle$ and β they should satisfy that:

$$\sum_h \langle\alpha|h\rangle\rangle\langle\langle h|\beta\rangle S_h^{h'} = n_{\alpha\beta}^{h'} \quad \textit{or equivalently} \quad \langle a|h'\rangle\rangle\langle\langle h'|b\rangle = \sum_h S_h^{h'} n_{ab}^{h}. \tag{7}$$

where $S_h^{h'}$ is the modular S matrix of the parent CFT and $n_{\alpha\beta}^{h'}$ are non-negative integers representing the multiplicities of the representation with highest weight h' in the open channel partition function $Z_{\alpha\beta}$.

A solution to the Cardy's condition is given by comparing it with the Verlinde formula. These solutions are called the Cardy States:

Definition 4. ***Cardy States***

Cardy states are linear combination of Ishibashi states with coefficients given by:

$$|a\rangle = \sum_j \frac{S_a^j}{(S_0^j)^{1/2}} |j\rangle\rangle \tag{8}$$

Proof: Taking n_{ab}^i as the fusion matrix N_{bc}^a and Cardy states into Cardy's condition, we can see that the Cardy's condition exactly gives out the Verlinde formula.

1.1.4 Elementary Boundary States

Looking at the Cardy's condition we might see that, if a state $|a\rangle$ satisfy the condition then $\mathbb{N}_+ |a\rangle$ must also satisfy this condition, for we only insist n_{ab}^i to be non-negative integers. However, we usually only consider the elementary boundary states that can not be decomposed into a sum of other boundary states. Thus we have the definition:

Definition 5. ***Elementary Boundary States***

For a parent CFT, a induced BCFT have following elementary boundary states $\{|a_i\rangle\}$ that shall satisfy the following normalization and orthogonality condition:

$$n_{a_i a_j}^0 = \delta_{ij} \tag{9}$$

The normalization is physical, this is because for a field theory, we assume to have a unique vaccum state, thus if $n_{ab}^0 \geq 2$ we will have multiple vacuum states in the open channel, which is unphysical, and shall be interpret as putting two theory togethor. Our goal is to find all elementary boundary states of a BCFT induced from a parent CFT. Moreover, we can see that:

Lemma 1. *Cardy states are elementary boundary states.*

This is because, cardy's state is a solution when considering $n_{ab}^i = N_{ab}^i$ which is the bulk fusion matrix. And we know that according to the 2 point function of bulk primary fields, the vacuum representation only appear once in the fusion of two primary fields when they are the same, written in operator algebra as:

$$\phi_a \times \phi_b = \mathbb{1}\delta_{ab} + \cdots \tag{10}$$

1.2 Ising CFT

Ising CFT as the simplist unitary minimal model $M(3/4)$ with central charge $c = 1/2$. Here I will briefly review some basic data of Ising CFT that will be used in the following sections. For a more detailed introduction to Ising CFT, one can refer to [3].

1.2.1 Ising CFT Data

Ising CFT is the simplest A-series unitary minimal model $M(3,4)$ with central charge $c = \frac{1}{2}$. It has three primary fields:

- Identity operator $\mathbb{1}$ with conformal weights $h_{\mathbb{1}} = \bar{h}_{\mathbb{1}} = 0$,
- Spin operator σ with conformal weights $h_\sigma = \bar{h}_\sigma = \frac{1}{16}$
- Energy operator ϵ with conformal weights $h_\epsilon = \bar{h}_\epsilon = \frac{1}{2}$.

They satisfy the folowing operator algebra (fusion rules):

$$\sigma \times \sigma = \mathbb{1} + \epsilon, \tag{11}$$
$$\sigma \times \epsilon = \sigma, \tag{12}$$
$$\epsilon \times \epsilon = \mathbb{1}. \tag{13}$$

1.2.2 Virasoro Characters of Ising CFT

Charaters of a Virasoro Representation is defined as :

Definition 6. ***Character***
The character $\chi_h(q)$ of a Virasoro representation with highest weight h is defined as:

$$\chi_h(q) = \mathrm{Tr}_{\mathcal{H}_h}\left(q^{L_0 - c/24}\right), \tag{14}$$

For Ising CFT the characters of the three primary fields are given by:

$$\chi_0(q) = \frac{1}{2} q^{-\frac{1}{48}} \left(\prod_{n=0}^{\infty} \left(1 + q^{\frac{1}{2}+n}\right) + \prod_{n=0}^{\infty} \left(1 - q^{\frac{1}{2}+n}\right) \right), \tag{15}$$

$$\chi_{1/2}(q) = \frac{1}{2} q^{-\frac{1}{48}} \left(\prod_{n=0}^{\infty} \left(1 + q^{\frac{1}{2}+n}\right) - \prod_{n=0}^{\infty} \left(1 - q^{\frac{1}{2}+n}\right) \right), \tag{16}$$

$$\chi_{1/16}(q) = q^{\frac{1}{24}} \prod_{n=1}^{\infty} (1 + q^n) \tag{17}$$

1.2.3 Modular S Matrix and Transformation of Characters

The modular S Matrix of Ising CFT is given by:

$$S^i_j = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \end{pmatrix} \tag{18}$$

with the column and row order $\mathbb{1}, \epsilon, \sigma$. The characters form a representation of the modular group with the transformation:

$$\chi_i(q) = \sum_j S^j_i \chi_j(\tilde{q}). \tag{19}$$

if we take $q = e^{-\frac{\pi\beta}{L}}$ and $\tilde{q} = e^{-\frac{4\pi L}{\beta}}$, we have:

$$\chi_0(\tilde{q}) = \frac{1}{2}\left(\chi_0(q) + \chi_{1/2}(q)\right) + \frac{1}{\sqrt{2}}\chi_{1/16}(q), \tag{20}$$

$$\chi_{1/2}(\tilde{q}) = \frac{1}{2}\left(\chi_0(q) + \chi_{1/2}(q)\right) - \frac{1}{\sqrt{2}}\chi_{1/16}(q), \tag{21}$$

$$\chi_{1/16}(\tilde{q}) = \frac{1}{\sqrt{2}}\left(\chi_0(q) - \chi_{1/2}(q)\right). \tag{22}$$

as well as the inverse transformation:

$$\chi_0(q) = \frac{1}{2}\left(\chi_0(\tilde{q}) + \chi_{1/2}(\tilde{q})\right) + \frac{1}{\sqrt{2}}\chi_{1/16}(\tilde{q}), \tag{23}$$

$$\chi_{1/2}(q) = \frac{1}{2}\left(\chi_0(\tilde{q}) + \chi_{1/2}(\tilde{q})\right) - \frac{1}{\sqrt{2}}\chi_{1/16}(\tilde{q}), \tag{24}$$

$$\chi_{1/16}(q) = \frac{1}{\sqrt{2}}\left(\chi_0(\tilde{q}) - \chi_{1/2}(\tilde{q})\right). \tag{25}$$

1.3 Ising BCFT

For a BCFT induced from Ising CFT, we can construct its boundary states by solving the Ishibashi condition and the Cardy's condition.

1.3.1 Ishibashi States of Ising BCFT

The Ishibashi states of Ising CFT can be constructed from its three primary states:

Definition 7. ***Ising Ishibashi States***
Ising CFT has three Ishibashi states:

$$|0\rangle\rangle = \sum_N |0;N\rangle \otimes \overline{|0;N\rangle}, \tag{26}$$

$$|1/2\rangle\rangle = \sum_N |1/2;N\rangle \otimes \overline{|1/2;N\rangle}, \tag{27}$$

$$|1/16\rangle\rangle = \sum_N |1/16;N\rangle \otimes \overline{|1/16;N\rangle}. \tag{28}$$

1.3.2 Cardy States of Ising BCFT

Using the modular S matrix of Ising CFT, we can write down the Cardy states by definition:

Definition 8. ***Ising Cardy States***
Ising BCFT exist three Cardy states:

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle\rangle + \frac{1}{\sqrt{2}}|1/2\rangle\rangle + \frac{1}{\sqrt[4]{2}}|1/16\rangle\rangle \tag{29}$$

$$|-\rangle = \frac{1}{\sqrt{2}}|0\rangle\rangle + \frac{1}{\sqrt{2}}|1/2\rangle\rangle - \frac{1}{\sqrt[4]{2}}|1/16\rangle\rangle \tag{30}$$

$$|f\rangle = |0\rangle\rangle - |1/2\rangle\rangle. \tag{31}$$

In Cardy's original paper [1], these three boundary states are argued to correspond to the three physical boundary conditions of the Ising model[1]:

- $|+\rangle$: fixed boundary condition with spins fixed to $+1$ at the boundary.
- $|-\rangle$: fixed boundary condition with spins fixed to -1 at the boundary.
- $|f\rangle$: free boundary condition with spins free to fluctuate at the boundary.

1.3.3 Partition Functions of Ising BCFT

With the Cardy states, we can compute the partition functions of Ising BCFT on a compatified strip (cylinder) with different boundary conditions:

Partition Function	Close Channel	Expression in $\tilde{q}$	Expression in q
$Z_{++} = Z_{--}$	$\langle \pm \| \tilde{q}^{\frac{1}{2}(L_0+\bar{L}_0-c/12)} \| \pm \rangle$	$\frac{1}{2}\chi_0(\tilde{q}) + \frac{1}{2}\chi_{1/2}(\tilde{q}) + \frac{1}{\sqrt{2}}\chi_{1/16}(\tilde{q})$	$\chi_0(q)$
$Z_{+-} = Z_{-+}$	$\langle \pm \| \tilde{q}^{\frac{1}{2}(L_0+\bar{L}_0-c/12)} \| \mp \rangle$	$\frac{1}{2}\chi_0(\tilde{q}) + \frac{1}{2}\chi_{1/2}(\tilde{q}) - \frac{1}{\sqrt{2}}\chi_{1/16}(\tilde{q})$	$\chi_{1/2}(q)$
$Z_{+f} = Z_{-f}$	$\langle \pm \| \tilde{q}^{\frac{1}{2}(L_0+\bar{L}_0-c/12)} \| f \rangle$	$\frac{1}{\sqrt{2}}\chi_0(\tilde{q}) - \frac{1}{\sqrt{2}}\chi_{1/2}(\tilde{q})$	$\chi_{1/16}(q)$
Z_{ff}	$\langle f \| \tilde{q}^{\frac{1}{2}(L_0+\bar{L}_0-c/12)} \| f \rangle$	$\chi_0(\tilde{q}) + \chi_{1/2}(\tilde{q})$	$\chi_0(q) + \chi_{1/2}(q)$

Table 1: Partition function and corresponding characters in both $\tilde{q}$ and q expansions. {tab::

We can see that the expression in q always have non-negative integer coefficients, which satisfy the Cardy condition.

1.4 Majorana CFT

A Majorana CFT (also known as free massless Majorana fermion CFT) is a two-dimensional conformal field theory describing free massless Majorana fermions. I'll briefly review the canonical quantization of Majorana CFT here.

1.4.1 Definition of Majorana CFT

A Majorana CFT is defined by the following action:

Definition 9. ***Majorana Fermion Action***
The action of a free Majorana fermion $\psi(z)$ and $\bar{\psi}(\bar{z})$ on a complex plane is given by:

$$S = \frac{1}{2\pi} \int d^2z \, \left(\psi \bar{\partial} \psi + \bar{\psi} \partial \bar{\psi} \right), \tag{32}$$

where $\partial = \frac{\partial}{\partial z}$ and $\bar{\partial} = \frac{\partial}{\partial \bar{z}}$ and $d^2z = d^2x = dxdy$.

The equations of motion derived from the action are:

$$\bar{\partial}\psi(z) = 0, \quad \partial\bar{\psi}(\bar{z}) = 0, \tag{33}$$

which means that $\psi(z)$ is a holomorphic field and $\bar{\psi}(\bar{z})$ is an anti-holomorphic field.

[1]This is an argument made by considering the transformation of the boundary states under $\mathbb{Z}_2$ symmetry of Ising CFT. It is not a rigorous theorem.

1.4.2 Canonical Quantization

To quantize the Majorana CFT, we impose radial quantization of the theory on a complex plane. The EoM gives us that the fields can be expanded as:

$$\psi(z) = \sum_k b_k z^{-k-1/2} \quad \bar{\psi}(\bar{z}) = \sum_k \bar{b}_k \bar{z}^{-k-1/2} \tag{34}$$

it is a common convention to take the $z^{1/2}$ in the mode expansion. The OPE of the fields gives us the anti-commutation relations of the modes:

$$\{b_k, b_l\} = \delta_{k+l,0}, \quad \{\bar{b}_k, \bar{b}_l\} = \delta_{k+l,0}, \quad \{b_k, \bar{b}_l\} = 0. \tag{35}$$

Moreover the Majorana fermion condition $\psi(z) = \psi^\dagger(z)$ gives us that the modes satisfy:

$$b_k^\dagger = b_{-k}, \quad \bar{b}_k^\dagger = \bar{b}_{-k}. \tag{36}$$

The Virasoro generators can be written in terms of the modes as:

$$L_n = \frac{1}{2}\sum_k (k + \frac{1}{2}) : b_{n-k} b_k : \tag{37}$$

1.4.3 Neveu-Schwarz and Ramond Sectors

We can impose two different boundary conditions on the fermion fields:

- **Neveu-Schwarz (NS) Boundary Condition**:

$$\psi(e^{2\pi i} z) = \psi(z), \quad \bar{\psi}(e^{-2\pi i}\bar{z}) = \bar{\psi}(\bar{z}). \tag{38}$$

  This boundary condition leads to half-integer moding of the fermion fields:

$$k \in \mathbb{Z} + \frac{1}{2}. \tag{39}$$

- **Ramond (R) Boundary Condition**:

$$\psi(e^{2\pi i} z) = -\psi(z), \quad \bar{\psi}(e^{-2\pi i}\bar{z}) = -\bar{\psi}(\bar{z}). \tag{40}$$

  This boundary condition leads to integer moding of the fermion fields:

$$k \in \mathbb{Z}. \tag{41}$$

Different boundary conditions lead to different mode expansions of the fermion fields as well as different Hilbert spaces of the theory. The Hilbert space of the theory is thus classified into two sectors: the NS sector and the R sector. The Hamiltonian of the Majorana CFT in both sectors can be written as:

Theorem 4. ***Majorana CFT Hamiltonian***

The Hamiltonian of the Majorana CFT on a complex plane is given by:

$$\begin{aligned}
H &= L_0 + \bar{L}_0 && (42)\\
L_0 &= \sum_{k>0} k\, b_{-k} b_k, \qquad \bar{L}_0 = \sum_{k>0} k\, \bar{b}_{-k}\bar{b}_k, \qquad (\mathrm{NS}:\ k \in \mathbb{Z} + \tfrac{1}{2}) && (43)\\
L_0 &= \sum_{k>0} k\, b_{-k} b_k + \frac{1}{16}, \qquad \bar{L}_0 = \sum_{k>0} k\, \bar{b}_{-k}\bar{b}_k + \frac{1}{16}, \qquad (\mathrm{R}:\ k \in \mathbb{Z}) && (44)
\end{aligned}$$

1.4.4 Zero Mode and Ground State Degeneracy

To construct the Hilbert space of the Majorana CFT, we need to define the ground states in both NS and R sectors. The NS sector works straightforwardly with a zero vacuum energy after normalization and a unique ground state $|0\rangle_{NS}$ defined by:

$$b_k |0\rangle_{NS} = 0, \quad \bar{b}_k |0\rangle_{NS} = 0, \quad k > 0. \tag{45}$$

However, the R sector is more subtle due to the presence of zero modes b_0 and $\bar{b}_0$. These zero modes satisfy the Clifford algebra:

$$\{b_0, b_0\} = 1, \quad \{\bar{b}_0, \bar{b}_0\} = 1, \quad \{b_0, \bar{b}_0\} = 0. \tag{46}$$

We can see that the zero modes b_0 and $\bar{b}_0$ generate a two-dimensional representation of the Clifford algebra. Thus the ground state of the R sector is doubly degenerate according to the following theorm:

Theorem 5. *Two dimensional Cliffoed Algebra Representation have non-trivial representation for at least two dimensions.*

We can define two ground states $|+\rangle_R$ and $|-\rangle_R$ and the act of pauli matrix on it is:

$$\sigma_x |\pm\rangle_{\mathrm{R}} = |\mp\rangle_{\mathrm{R}}, \quad \sigma_y |\pm\rangle_{\mathrm{R}} = \mp i\, |\mp\rangle_{\mathrm{R}}, \quad \sigma_z |\pm\rangle_{\mathrm{R}} = \pm |\pm\rangle_{\mathrm{R}}. \tag{47}$$

And then we can represent the zero modes and the fermion parity operator as:

$$b_0 = \frac{1}{2} (\sigma_x + \sigma_y)\, (-1)^{\sum_{n>0} b_{-n} b_n + \bar{b}_{-n} \bar{b}_n}, \tag{48}$$

$$\bar{b}_0 = \frac{1}{2} (\sigma_x - \sigma_y)\, (-1)^{\sum_{n>0} b_{-n} b_n + \bar{b}_{-n} \bar{b}_n}, \tag{49}$$

$$\tag{50}$$

For the ground states, we have:

$$b_0 |\pm\rangle_{\mathrm{R}} = \frac{1}{\sqrt{2}} e^{\pm i\pi/4} |\mp\rangle_{\mathrm{R}}, \quad \bar{b}_0 |\pm\rangle_{\mathrm{R}} = \frac{1}{\sqrt{2}} e^{\mp i\pi/4} |\mp\rangle_{\mathrm{R}}. \tag{51}$$ {eq:Ma

1.4.5 Fermion Parity

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We hope to define (in many cases this operator might not be well defined) a fermion parity operator $(-1)^F$ in a Fermionic CFT.

Definition 10. ***Fermion Parity Operator***
This operator counts the parity of fermion number in the theory, satisfying:

$$\{(-1)^F, b_k\} = 0, \quad \{(-1)^F, \bar{b}_k\} = 0, \quad ((-1)^F)^2 = 1. \tag{52}$$

In the NS sector, the fermion parity operator is explicitly given by:

$$(-1)^F = (-1)^{\sum_{k>0} b_{-k} b_k + \sum_{k>0} \bar{b}_{-k} \bar{b}_k}. \tag{53}$$

In the R sector, there are two different ways of defining the fermion number operator:

$$(-1)^F = \begin{cases} -2i\, b_0 \bar{b}_0\, (-1)^{F_{nz}}, \\ +2i\, b_0 \bar{b}_0\, (-1)^{F_{nz}}. \end{cases} \tag{54}$$

where $(-1)^{F_{nz}} = (-1)^{\sum_{k>0} b_{-k} b_k + \sum_{k>0} \bar{b}_{-k} \bar{b}_k}$. In this note, we will use the first definition of fermion parity operator in the R sector.

1.4.6 Hilbert Space Structure

With the ground state structure and mode expansion, we can construct the Hilbert space of the Majorana CFT. The Hilbert space is classified into two sectors: the NS sector and the R sector.

- **NS Sector Hilbert Space**: The Hilbert space in the NS sector is constructed by acting the creation operators $b_{-k}, \bar{b}_{-k}$ ($k > 0$ and $k \in \mathbb{Z} + 1/2$) on the unique ground state $|0\rangle_{NS}$:

$$\mathcal{H}_{NS} = \text{span}\left\{ b_{-k_1} b_{-k_2} \cdots \bar{b}_{-l_1} \bar{b}_{-l_2} \cdots |0\rangle_{NS} \mid k_i, l_j > 0, k_i, l_j \in \mathbb{Z} + \frac{1}{2} \right\}. \tag{55}$$

- **R Sector Hilbert Space**: The Hilbert space in the R sector is constructed by acting the creation operators $b_{-k}, \bar{b}_{-k}$ ($k > 0$ and $k \in \mathbb{Z}$) on the doubly degenerate ground states $|\pm\rangle_R$:

$$\mathcal{H}_R = \text{span}\left\{ b_{-k_1} b_{-k_2} \cdots \bar{b}_{-l_1} \bar{b}_{-l_2} \cdots |\pm\rangle_R \mid k_i, l_j > 0, k_i, l_j \in \mathbb{Z} \right\}. \tag{56}$$

2 Majorana BCFT

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As a standard approach to boundary conformal field theory, we consider a Majorana CFT defined on a compactified strip with two boundaries (open channel), or equivalently on a cylinder (closed channel). We compute the partition function in both channels and present explicit expressions for the corresponding boundary states. Although there exist several references on this subject [6, 2, 4, 5], none of them provides a fully explicit and correct derivation of the partition function and the boundary states.

2.1 Boundary Conditions of Majorana CFT

It is well know that the Majorana CFT on the UHP admits two kinds of conformal boundary conditions:

- **Free** (+) **Boundary Condition:** $\psi = \bar{\psi}$ at the boundary.

- **Fixed** ($-$) **Boundary Condition:** $\psi = -\bar{\psi}$ at the boundary.

It is belived that the free boundary condition corresponds to free spin boundary condition of Ising model while the fixed boundary condition corresponds to fixed spin boundary condition of Ising model.

We can consistently define these two boundary conditions on any manifold with boundaries. And if we focus on our main topic: Majorana CFT on a compatified strip with two boundaries. We consider the strip $\mathcal{S} = \{z = x + iy | x \in [0, L], y \in \mathbb{R}\}$ with two boundaries at $x = 0$ and $x = L$. Then strip is compatified by identifying $y \sim y + \beta$.

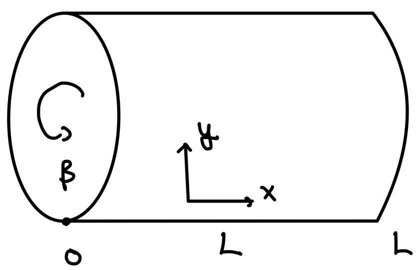

Figure 1: Compatified strip with two boundaries at $x = 0$ and $x = L$, compatified by identifying $y \sim y + \beta$.

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On this manifold, we can write down the boundary conditions at $x = 0$ and $x = L$ as:

- **Free (+) Boundary Condition:**

$$\psi(0,y) = i\bar{\psi}(0,y), \quad \psi(L,y) = -i\bar{\psi}(L,y) \tag{57}$$

- **Fixed (−) Boundary Condition:**

$$\psi(0,y) = -i\bar{\psi}(0,y), \quad \psi(L,y) = i\bar{\psi}(L,y) \tag{58}$$

2.2 Open Channel Partition Function

With the boundary conditions in hand we can canonically quantize the Majorana CFT on this compatified strip and get the open channel partition function.

2.2.1 Doubling Trick

We aren't familiar with the mode expansion on a strip and quantization on a strip, however, Majorana CFT on a cylinder is well studied. Thus we may use a trick to double a strip and get a cylinder. This is the so-called doubling trick. We can write the two boundary conditions in a compact form:

$$\psi(0,y) + ai\bar{\psi}(0,y) = 0 \quad \psi(L,y) - bi\bar{\psi}(L,y) = 0, \tag{59}$$

- when $a = b = 1$ we have the fixed (−) boundary condition at both boundaries
- when $a = b = -1$ we have the free (+) boundary condition at both boundaries
- when $a = -b = 1$ we have the mixed boundary condition: fixed at $x = 0$ and free at $x = L$
- when $a = -b = -1$ we have the mixed boundary condition: free at $x = 0$ and fixed at $x = L$

then we define a anxilary field $\bar{\psi}_0(x,y) = -a\bar{\psi}(x,y)$. Thus the boundary condition can be written as:

$$\psi(0,y) = i\bar{\psi}_0(0,y) \quad \psi(L,y) = -i\frac{b}{a}\bar{\psi}_0(L,y) \tag{60}$$

Then we can extend the field $\psi(x,y)$ to the region $x \in [-L, L]$ by defining:

Theorem 6. ***Doubled Theory***
We define a field on the strip $[-L, L] \times \mathbb{R}$ with:

$$\Psi(x,y) = \begin{cases} \psi(x,y) & \text{if} \quad x \in [0,L] \\ i\bar{\psi}_0(-x,y) & \text{if} \quad x \in [-L,0] \end{cases} \tag{61}$$

The original action can be written as:

$$S^S = \frac{1}{2\pi}\int_0^L dx \int_{-\infty}^{\infty} dy \; [\psi\bar{\partial}\psi + \bar{\psi}\partial\bar{\psi}] = S^C = \frac{1}{2\pi}\int_{-L}^{L} dx \int_{-\infty}^{\infty} dy \; \Psi\bar{\partial}\Psi = \frac{1}{2\pi}\int_z \Psi\bar{\partial}\Psi \tag{62}$$

where $z = x + iy$.

We can see that at the $x = 0$ boundary, the field $\Psi(x,y)$ is continuous. While at the $x = L$ boundary, we have:

$$\Psi(L,y) = -\frac{b}{a}\Psi(-L,y) \tag{63}$$

Thus we can understand the theory of two decoupled field $\psi(x,y)$ and $\bar{\psi}(x,y)$ on a strip with boundary conditions as a theory of a single ifield $\Psi(x,y)$ on a circle of circumference $2L$.

- Same boundary conditions $(+,+)$ or $(-,-)$ correspond to the Neveu-Schwarz (NS) sector with anti-periodic boundary condition: $\Psi(L,y) = -\Psi(-L,y)$
- Different boundary conditions $(+,-)$ or $(-,+)$ correspond to the Ramond (R) sector with periodic boundary condition: $\Psi(L,y) = \Psi(-L,y)$

Eventually, we learn that the Majorana CFT on a strip is equivalent to a Chiral Majorana CFT on a cylinder. We then can perform the quantization on a cylinder to get the quantized theory.

2.2.2 Canonical Quantization

The canonical quantization of Majoraca CFT on a cylinder is well studied. The Hamiltonian of the theory is given by:

Theorem 7. ***Chiral Majorana CFT Hamiltonian on a Cylinder***
The Hamiltonian of the theory is:

$$H_{open} = \frac{\pi}{L}\left(\sum_{k>0} kb_{-k}b_k + E_0\right) \tag{64}$$ {eq:Ma

where b_k are the mode operators with $k \in \mathbb{Z} + \frac{1}{2}$ for NS sector and $k \in \mathbb{Z}$ for R sector, satisfying the anti-commutation relation and E_0 is the ground state energy:

$$E_0 = \begin{cases} -\frac{1}{48} & \text{NS sector} \\ \frac{1}{24} & \text{R sector} \end{cases} \tag{65}$$

2.2.3 Zero Mode and Ground State Degeneracy(?)

With a chiral Majorana fermion on a cylinder, when considering the Ramond sector, we know that the zero mode only has one-dimensional Clifford algebra:

$$\{b_0, b_0\} = 1 \quad b_0^2 = \frac{1}{2} \tag{66}$$

Naively, we may think that we only have one ground state $|0\rangle_R$ satisfying $b_k|0\rangle = 0$ for $k > 0$. The zero mode b_0 then acts on the ground state as:

$$b_0 = \frac{1}{\sqrt{2}}|0\rangle_R \quad \text{or} \quad b_0 = -\frac{1}{\sqrt{2}}|0\rangle_R \tag{67}$$

However, things get worse when we try to define the fermion parity operator $(-1)^F$. This is due to a mathematical theorem:

Theorem 8. ***Odd Dimensional Clifford Algebra Grading***
Clifford algebra with odd number of generators doesn't admit a $\mathbb{Z}_2$ grading, ie there is no way to define a fermion parity operator $(-1)^F$ that anti-commutes with all the generators.

2.2.4 Boundary Fermion

However, we **manually wish to define** a fermion parity operator $(-1)^F$ and **we wish to** have a ground state degeneracy. To achieve this, we introduce an auxilary boundary free Majorana fermion $\xi(t)$ with dynamical terms only on the boundary of one of the boundary conditions.

Theorem 9. ***Boundary Fermion***

We introduce an auxilary boundary free Majorana fermion $\xi(t)$ with dynamical terms only on the boundary of one of the boundary conditions. we have two choices:

- *Add $\xi(t)$ on the free boundary condition boundary (we will focus on this choice in this note)*
- *Add $\xi(t)$ on the fixed boundary condition boundary*

The action of the boundary fermion is:

$$S_\xi = \frac{i}{4}\int dt\ \xi(t)\frac{d}{dt}\xi(t) \tag{68}$$

Under canonical quantization, the boundary fermion satisfies the anti-commutation relation:

$$\{\xi(t), \xi(t')\} = 2\delta(t - t') \tag{69}$$

This defines a extra boundary Hilbert space $\mathcal{H}_B = Span\{|0\rangle_B, \xi|0\rangle_B\}$.

With the inclusion of boundary fermions, the ground-state degeneracy is restored and we can define the fermion parity operator properly.

$(-,-)$ **Boundary Conditions.** When both boundaries carry the same fixed boundary conditions, no boundary fermion is introduced. The bulk Majorana fermion has no zero mode in this case, and thus there is no ground-state degeneracy. We can define the fermion parity operator as:

$$(-1)^F = (-1)^{F_{\text{nz}}} = (-1)^{\sum_{k>0} b_{-k}b_k}. \tag{70}$$

$(+,-),(-,+)$ **Boundary Conditions.** When the two boundaries carry different boundary conditions, a single boundary fermion $\xi(t)$ exists on the free boundary. It provides an additional zero mode satisfying

$$\{\xi,\xi\} = 2, \qquad \xi^2 = 2, \qquad \{\xi, b_k\} = 0 \quad \forall\, k. \tag{71}$$

Together with the bulk zero mode b_0, ξ generates a Clifford algebra, leading to a degenerate ground state. We can define the fermion parity operator as:

$$(-1)^F = \sqrt{2}\, i\, \xi b_0\, (-1)^{F_{\text{nz}}}. \tag{72}$$

$(+,+)$**Boundary Conditions.** When both boundaries are free, two boundary fermions $\xi_1(t)$ and $\xi_2(t)$ are present. In this case the bulk Majorana fermion has no zero mode. Nevertheless, the boundary fermions satisfy

$$\{\xi_i,\xi_j\} = 2\delta_{ij}, \qquad \xi_i^2 = 2, \qquad \{\xi_i, b_k\} = 0 \quad \forall\, k, \tag{73}$$

and generate a two-dimensional Clifford algebra, which again implies a ground-state degeneracy. We can define the fermion parity operator as:

$$(-1)^F = i\, \xi_1\xi_2\, (-1)^{F_{\text{nz}}}. \tag{74}$$

2.2.5 Open Channel Hilbert Space

The Hilbert spaces associated with the different boundary conditions are summarized as follows.

- $(-,-)$ **boundary conditions.** The Hilbert space is generated by acting with the negative modes b_{-k} ($k \in \mathbb{Z}+\frac{1}{2}$) on the NS vacuum $|0\rangle_{\text{NS}}$.
- $(+,-)$ **and** $(-,+)$ **boundary conditions.** The Hilbert space is constructed by acting with the negative modes b_{-k} ($k \in \mathbb{Z}_{>0}$) on the degenerate ground states.
- $(+,+)$ **boundary conditions.** The Hilbert space is generated by acting with the negative modes b_{-k} ($k \in \mathbb{Z}+\frac{1}{2}$) on the degenerate ground states. The resulting Hilbert space is twice as large as that of the $(-,-)$ sector.

2.2.6 Open Channel Partition Functions

Then we can finally calculate the partition functions in the open channel. The partition function is defined as:

> **Definition 11.** ***Open Channel Partition Function***
> *The open channel partition function is defined as:*
>
> $$Z = \mathrm{Tr}_{\mathcal{H}}\left[e^{-\beta H}\right] \tag{75}$$
>
> *Yet for fermions we can compatify with periodic or anti-periodic boundary condition, thus we have 2 different partition functions:*
>
> $$Z_{-} = \mathrm{Tr}_{\mathcal{H}}\left[(-1)^F e^{-\beta H}\right] \quad Z_{+} = \mathrm{Tr}_{\mathcal{H}}\left[e^{-\beta H}\right] \tag{76}$$

We then calculate the partition functions for the 4 types of boundary conditions, and explicitly write them in terms of Ising characters in terms of $q = e^{-\pi\beta/L}$ and $\tilde{q} = e^{-4\pi L/\beta}$:

Partition Function	q character	$\tilde{q}$ character
$Z_{(--),-}$	$\chi_0(q) - \chi_{1/2}(q)$	$\sqrt{2}\,\chi_{1/16}(\tilde{q})$
$Z_{(--),+}$	$\chi_0(q) + \chi_{1/2}(q)$	$\chi_0(\tilde{q}) + \chi_{1/2}(\tilde{q})$
$Z_{(++),-}$	0	0
$Z_{(++),+}$	$2\big(\chi_0(q) + \chi_{1/2}(q)\big)$	$2\big(\chi_0(\tilde{q}) + \chi_{1/2}(\tilde{q})\big)$
$Z_{(-+),-}$	0	0
$Z_{(-+),+}$	$2\,\chi_{1/16}(q)$	$\sqrt{2}\big(\chi_0(\tilde{q}) - \chi_{1/2}(\tilde{q})\big)$

Table 2: Free fermion open-channel partition functions in both q and $\tilde{q}$ characters. The notation $Z_{(\pm,\pm),\pm}$, $(\pm,\pm)$ in the braket stands for boundary conditions while $,\pm$ outside the braket stands for the insertion of fermion parity operator $(-1)^F$ in the partition function. {tab:c

2.3 Closed Channel Partition Function

After the calculation of the open channel partition function, we want to look for boundary states that correspond to the 2 types of boundary conditions we have discussed in the open channel.

2.3.1 Fermionic Boundary State Formalism

We have seen that every choice of boundary conditions in the open channel corresponds to two partition functions in the closed channel. Thus, we have to first generalize the boundary state formalism to Fermionic CFT.

Definition 12. ***Fermionic Boundary State***

We define the boundary state $|\alpha, NS\rangle \in \mathcal{H}_{NS}$ *and* $|\alpha, R\rangle \in \mathcal{H}_R$ *in Fermionic CFT as the state that satisfies:*

$$\langle\alpha, NS|\, e^{-L_1 H_{\text{closed}}}\, |\beta, NS\rangle \;=\; \text{Tr}_{\mathcal{H}_{\alpha\beta}}\big(e^{-L_2 H_{\text{open}}}\big) = Z_{(\alpha\beta),+}, \tag{77}$$

$$\langle\alpha, R|\, e^{-L_1 H_{\text{closed}}}\, |\beta, \text{R}\rangle \;=\; \text{Tr}_{\mathcal{H}_{\alpha\beta}}\big((-1)^F e^{-L_2 H_{\text{open}}}\big) = Z_{(\alpha\beta),-}. \tag{78}$$

Thus we have two boundary states $|\alpha, NS\rangle$ and $|\alpha, R\rangle$ living in the Hilbert space of both sectors corresponding to a single boundary condition α

2.3.2 Canonical Quantization in Closed Channel

The closed channel is the parent Majorana CFT on a cylinder with circumference β and length L. The quantization result is well known. We will list out the Hamiltonian:

Theorem 10. ***Majorana CFT Hamiltonian on a Cylinder***

$$H_{closed} = \frac{2\pi}{\beta}\left(2E_0 + \sum_{k>0} k\left(a_{-k}a_k + \bar{a}_{-k}\bar{a}_k\right)\right) \quad E_0 = \begin{cases} -\frac{1}{48} & \text{NS sector} \\ \frac{1}{24} & \text{R sector} \end{cases} \tag{79}$$

And we define the fermion parity operator as:

$$(-1)^F = \begin{cases} (-1)^{F_{\text{nz}}} & \text{NS sector} \\ -2ia_0\bar{a}_0(-1)^{F_{\text{nz}}} & \text{R sector} \end{cases} \tag{80}$$

Here $F_{\text{nz}} = \sum_{k>0} a_{-k}a_k + \sum_{k>0} \bar{a}_{-k}\bar{a}_k$ is the non-zero mode fermion number operator.

2.3.3 Gluing Conditions and Ishibashi States

Boundary conditions shall turn into gluing condition for boundary states in the closed channel. With consistency considerations, we can derive the gluing conditions for the Majorana CFT on a cylinder as:

- **Free** $(+)$ **Boundary Condition:** for a boundary state on a boundary with free $(+)$ boundary condition, it shall satisfy the following gluing condition:

$$(a_k - i\bar{a}_{-k})\,|+, NS/R\rangle = 0\ , \qquad \langle +, NS/R|\,(a_{-k} + i\bar{a}_k) = 0\ , \tag{81}$$

- **Fixed** $(-)$ **Boundary Condition:** for a boundary state on a boundary with fixed $(-)$ boundary condition, it shall satisfy the following gluing condition:

$$(a_k + i\bar{a}_{-k})\,|-, NS/R\rangle = 0\ , \qquad \langle -, NS/R|\,(a_{-k} - i\bar{a}_k) = 0\ , \tag{82}$$

For these gluing condition, it is able to construct a set of Ishibashi states that satisfy them:

- **NS Sector Ishibashi States:**

$$|\pm, NS\rangle\rangle = \prod_{k\in\mathbb{N}_+-1/2} e^{i\pm a_{-k}\bar{a}_{-k}}|0\rangle_{\text{NS}} \tag{83}$$

  The $+$ state is solution to the gluing condition of free boundary condition while the $-$ state is solution to the gluing condition of fixed boundary condition.

- **R Sector Ishibashi States:**

$$|\pm, R\rangle\rangle = \prod_{k\in\mathbb{N}_+} e^{i\pm a_{-k}\bar{a}_{-k}}|\pm\rangle_{\mathrm{R}} \tag{84}$$

  The $+$ state is solution to the gluing condition of free boundary condition while the $-$ state is solution to the gluing condition of fixed boundary condition. And $|\pm\rangle_R$ are the degenerate R ground states

2.3.4 Boundary States and Partition Functions

We know that boundary states shall be linear combinations of the Ishibashi states we have constructed. However, we notice that for every boundary condition only exists one corresponding Ishibashi state. Thus, the boundary states are merely the Ishibashi states themselves up to a normalization factor

$$|+, NS\rangle \;\propto\; |\pm, NS\rangle\rangle, \qquad |-, NS\rangle \;\propto\; |-, NS\rangle\rangle, \tag{85}$$
$$|+, R\rangle \;\propto\; |+, R\rangle\rangle, \qquad |-, R\rangle \;\propto\; |-, R\rangle\rangle. \tag{86}$$

We can fix the normalization factors by comparing the closed channel partition functions with the open channel partition functions we have calculated before. We finally can get the boundary states as:

$$|-, NS\rangle = |-, NS\rangle\rangle \tag{87}$$
$$|-, R\rangle = \sqrt[4]{2}|-, R\rangle\rangle, \tag{88}$$
$$|+, NS\rangle = \sqrt{2}|+, NS\rangle\rangle, \tag{89}$$
$$|+, R\rangle = 0|+, R\rangle\rangle = 0. \tag{90}$$

With the closed channel partiton function listed out in the table below:

Open Channel	q character	$\tilde{q}$ character	Closed Channel
$Z_{(--),-}$	$\chi_0(q) - \chi_{1/2}(q)$	$\sqrt{2}\,\chi_{1/16}(\tilde{q})$	$\langle -, R\|e^{-LH_{\mathrm{closed}}}\|-, R\rangle$
$Z_{(--),+}$	$\chi_0(q) + \chi_{1/2}(q)$	$\chi_0(\tilde{q}) + \chi_{1/2}(\tilde{q})$	$\langle -, NS\|e^{-LH_{\mathrm{closed}}}\|-, NS\rangle$
$Z_{(++),-}$	0	0	$\langle +, R\|e^{-LH_{\mathrm{closed}}}\|+, R\rangle$
$Z_{(++),+}$	$2\big(\chi_0(q) + \chi_{1/2}(q)\big)$	$2\big(\chi_0(\tilde{q}) + \chi_{1/2}(\tilde{q})\big)$	$\langle +, NS\|e^{-LH_{\mathrm{closed}}}\|+, NS\rangle$
$Z_{(-+),-}$	0	0	$\langle -, R\|e^{-LH_{\mathrm{closed}}}\|+, R\rangle$
$Z_{(-+),+}$	$2\,\chi_{1/16}(q)$	$\sqrt{2}\big(\chi_0(\tilde{q}) - \chi_{1/2}(\tilde{q})\big)$	$\langle -, NS\|e^{-LH_{\mathrm{closed}}}\|+, NS\rangle$

Table 3: Free fermion open-channel partition functions in both q and $\tilde{q}$ characters, with the corresponding closed-channel overlaps. {tab:c

3 Bosonization of Majorana BCFT

{sec:B

In this section, we will discuss how to bosonize the Majorana CFT with boundaries in closed channel. This topic has been discussed in several references, [5, 4]. Yet none of them gives a complete and clear explanation of the process.

3.1 Bosonization Procedure

We will first make it clear by stating what we mean by bosonization and how to do it.

3.1.1 Bosonization of Fermionic Theories

Bosonization is transforming a fermionic theory into a bosonic theory. A naive idea of changing a fermionic theory into a bosonic theory is to simply bind all fermions in pairs. When two fermions are paired, they can behave as a boson. This naive idea works well, and mathematically it can be described as the GSO projection in fermionic theory.

Definition 13. ***GSO Projection***
The GSO projection can be understood as a projection onto the subspace of states with even fermion parity. For a fermionic theory with a well defined fermion parity operator $(-1)^F$, the GSO-projected partition function is defined as

$$Z_{\text{GSO}} = \mathrm{Tr}_{\mathcal{H}}\left[\frac{1+(-1)^F}{2}\, e^{-\beta H}\right]. \tag{91}$$

Equivalently, one may introduce the projection operator

$$P_{\text{GSO}} = \frac{1+(-1)^F}{2}, \tag{92}$$

which annihilates states of odd fermion parity and acts trivially on states of even fermion parity:

$$P_{\text{GSO}}\,|\text{odd}\rangle = 0, \qquad P_{\text{GSO}}\,|\text{even}\rangle = |\text{even}\rangle\,. \tag{93}$$

Thus, we can say that by bosonization we mean performing a GSO projection of the partition function and Hilbert space of the fermionic theory which turns out to be a partition funtion and Hilbert space of a bosonic theory.

3.1.2 Bosonization of Boundary Conditions/States

When it comes to a boundary theory, we also have to "bosonize" the boundary conditions. For a abstract boundary condition, it might be hard to understand how to "bosonize" it. However, the closed channel provides a perfect platform of underingstanding this by representing the boundary conditions as a boundary state, which can be practically GSO projected. The bosonization of boundary states can be defined as following two steps:

- First, GSO project the boundary states of the fermionic theory. We only keep the boundary states with even fermion parity.
- Choose the Linear combination of GSO projected boundary states that satisfies the Cardy's condition and are elementary boundary states. To make it a valid bosonic boundary state.

3.2 Bosonization of Majorana BCFT

We apply the above bosonization procedure to the Majorana BCFT. We first project the boundary states we obtained in the previous section. We may find that the three non-zero boundary states are all of even fermion parity, thus the GSO projection is trivial:

$$P_{\text{GSO}}|\pm, NS\rangle = |\pm, NS\rangle \quad P_{\text{GSO}}\,|-, R\rangle = |-, R\rangle \tag{94}$$

Then we want to find a linear combination of these GSO projected boundary states that satisfy the Cardy's condition and are elementary boundary states. Generally, it is hard to find a solution.

However, if we assume that the bosonized CFT is a diagonal CFT, then we can get a following solution:

$$|f\rangle = \frac{1}{\sqrt{2}} |+, NS\rangle \oplus 0 \tag{95}$$

$$|+\rangle = \frac{1}{\sqrt{2}} (|-, NS\rangle \oplus |-, R\rangle) \tag{96}$$

$$|-\rangle = \frac{1}{\sqrt{2}} (|-, NS\rangle \oplus - |-, R\rangle) \tag{97}$$

Then we can calculate the partition functions of these bosonized boundary states with:

$$e^{-LH_{\text{closed}}} = e^{-LH_{\text{closedNS}}} \oplus e^{-LH_{\text{closedR}}} \tag{98}$$

We list the results in the table below:

q character	$\tilde{q}$ character	Closed Channel
$\chi_0(q)$	$\frac{1}{2}\big(\chi_0(\tilde{q}) + \chi_{1/2}(\tilde{q})\big) + \frac{1}{\sqrt{2}}\chi_{1/16}(\tilde{q})$	$\langle +\lvert e^{-LH_{\text{closed}}}\rvert +\rangle$
$\chi_0(q)$	$\frac{1}{2}\big(\chi_0(\tilde{q}) + \chi_{1/2}(\tilde{q})\big) + \frac{1}{\sqrt{2}}\chi_{1/16}(\tilde{q})$	$\langle -\lvert e^{-LH_{\text{closed}}}\rvert -\rangle$
$\chi_{1/2}(q)$	$\frac{1}{2}\big(\chi_0(\tilde{q}) + \chi_{1/2}(\tilde{q})\big) + \frac{1}{\sqrt{2}}\chi_{1/16}(\tilde{q})$	$\langle -\lvert e^{-LH_{\text{closed}}}\rvert +\rangle$
$\chi_0(q) + \chi_{1/2}(q)$	$\chi_0(\tilde{q}) + \chi_{1/2}(\tilde{q})$	$\langle f\lvert e^{-LH_{\text{closed}}}\rvert f\rangle$
$\chi_{1/16}(q)$	$\frac{1}{\sqrt{2}}\big(\chi_0(\tilde{q}) - \chi_{1/2}(\tilde{q})\big)$	$\langle +\lvert e^{-LH_{\text{closed}}}\rvert f\rangle$
$\chi_{1/16}(q)$	$\frac{1}{\sqrt{2}}\big(\chi_0(\tilde{q}) - \chi_{1/2}(\tilde{q})\big)$	$\langle -\lvert e^{-LH_{\text{closed}}}\rvert f\rangle$

Table 4: Partition function of the bosonized boundary states.

We compare the above table with the known Ising CFT boundary states partition functions table 1 and we find they are exactly the same! Thus we identify the bosonized theory as the Ising BCFT and the bosonized boundary states as the Ising boundary states:

$$|f\rangle = |f\rangle_{\text{Ising}} \tag{99}$$

$$|+\rangle = |+\rangle_{\text{Ising}} \tag{100}$$

$$|-\rangle = |-\rangle_{\text{Ising}} \tag{101}$$

References

[1] John L. Cardy. Boundary conditions, fusion rules and the Verlinde formula. *Nuclear Physics B*, 324(3):581–596, October 1989.

[2] R. Chatterjee. Exact Partition Function and Boundary State of Critical Ising Model with Boundary Magnetic Field. *Modern Physics Letters A*, 10(12):973–984, April 1995.

[3] Philippe Di Francesco, Pierre Mathieu, and David Sénéchal. *Conformal Field Theory*. Graduate Texts in Contemporary Physics. Springer New York, New York, NY, 1997.

[4] Hiromi Ebisu and Masataka Watanabe. Fermionization of conformal boundary states. *Physical Review B*, 104(19):195124, November 2021.

[5] Kentaro Hori. Notes on Bosonization with Boundary. *Progress of Theoretical Physics Supplement*, 177:42–53, 2009.

[6] Rafael I. Nepomechie. Consistent superconformal boundary states. *Journal of Physics A: Mathematical and General*, 34(33):6509–6524, August 2001.

[7] Christian Northe. Young Researchers School 2024 Maynooth: Lectures on CFT, BCFT and DCFT, January 2025.

[8] Andreas Recknagel and Volker Schomerus. Boundary Conformal Field Theory and the Worldsheet Approach to D-Branes.